Journal of Experimental Marine Biology and Ecology 400 (2011) 218–226

Contents lists available at ScienceDirect

Journal of Experimental Marine Biology and Ecology

journal homepage: www.elsevier.com/locate/jembe

Community ecology in a warming world: The influence of temperature on interspecific interactions in marine systems

Rebecca L. Kordas [a,*], Christopher D.G. Harley [a], Mary I. O'Connor [a,b]

[a] Department of Zoology, University of British Columbia, Vancouver, BC, V6T 1Z4, Canada
[b] National Center for Ecological Analysis and Synthesis, 735 State St, Suite 300, Santa Barbara, CA 93101, United States

ARTICLE INFO

Keywords:
Species interaction
Temperature
Climate Change Ecology
Community Ecology
Metabolic Ecology

ABSTRACT

Ecological patterns are determined by the interplay between abiotic factors and interactions among species. As the Earth's climate warms, interactions such as competition, predation, and mutualism are changing due to shifts in per capita interaction strength and the relative abundance of interacting species. Changes in interspecific relationships, in turn, can drive important local-scale changes in community dynamics, biodiversity, and ecosystem functioning, and can potentially alter large-scale patterns of distribution and abundance. In many cases, the importance of indirect effects of warming, mediated by changing species interactions, will be greater—albeit less well understood—than direct effects in determining the community- and ecosystem-level outcomes of global climate change. Despite considerable community-specific idiosyncrasy, ecological theory and a growing body of data suggest that certain general trends are emerging at local scales: positive interactions tend to become more prevalent with warming, and top trophic levels are disproportionately vulnerable. In addition, important ecological changes result when the geographic overlap between species changes, and when the seasonal timing of life history events of interacting species falls into or out of synchrony. We assess the degree to which such changes are predictable, and urge advancement on several high priority questions surrounding the relationships between temperature and community ecology. An improved understanding of how assemblages of multiple, interacting species will respond to climate change is imperative if we hope to effectively prepare for and adapt to its effects.

Crown Copyright © 2011 Published by Elsevier B.V. All rights reserved.

Contents

1. Introduction . . . 218
2. The biological importance of temperature . . . 219
3. Interspecific variation in thermal sensitivity . . . 220
4. Incorporating time and space: phenology and biogeography. . . . 222
5. The search for generality . . . 223
6. Future research priorities . . . 224
Acknowledgements . . . 224
References . . . 225

1. Introduction

Temperature is one of the most fundamental determinants of biological patterns and processes. Many decades of laboratory-based research have demonstrated that variation in temperature has important and easily measured effects on biochemical and physiological rates. Because biochemical and physiological rates translate into organismal survival, growth, and reproduction, environmental temperature plays a large role in determining when and where species—particularly ectothermic species—can survive and thrive (Wethey, 1983; Thomas et al., 2000; Hochachka and Somero, 2002). Indeed, variation in temperature explains much of the spatial and temporal patterns we observe in the distribution and abundance of species around the world (Hutchins, 1947).

Although long recognized as biologically important, environmental temperature is currently being addressed with renewed vigor as anthropogenic climate change alters patterns of mean and extreme temperatures across the globe. Climate models suggest that the average

* Corresponding author. Tel.: +1 778 862 2000; fax: +1 604 822 2416.
E-mail address: kordas@zoology.ubc.ca (R.L. Kordas).

0022-0981/$ – see front matter. Crown Copyright © 2011 Published by Elsevier B.V. All rights reserved.
doi:10.1016/j.jembe.2011.02.029

temperature of the surface of the earth will warm by 1.7–4.4 °C by the end of the current century, with increases in mean temperatures and in the frequency and magnitude of extreme temperature events (IPCC, 2007). The magnitude of these projected changes varies from place to place (see Fig. 1). The broad-brush effects of warming are already observable across a wide variety of systems and taxa, with shifts in the distribution and abundance of species and the timing of life history events occurring largely as one would predict over spatial (e.g. latitudinal and altitudinal) and temporal (e.g., seasonal) thermal gradients (Sagarin et al., 1999; Parmesan and Yohe, 2003; Southward et al., 1995, 2005; Helmuth et al., 2006a; Mieszkowska et al., 2007). However, not every species has responded as predicted (e.g. Hawkins et al., 2009), and for the vast majority of species little to no data on responses to temperature exist. To better understand which species are shifting and why, and the ecological impacts of temperature changes of different magnitudes, tests of climate impacts must link processes from the climatological and biophysical to the physiological and demographic to produce a more refined understanding of how environmental temperature influences body temperature and thereby the distribution and abundance of species (Helmuth, 2009).

It has long been known, however, that temperature is not the sole determinant of where a species can live and how well it will perform. For example, Darwin (1959) recognized that many distributional patterns across thermal gradients seemed to depend more on interactions among species than upon the direct effects of temperature, an observation that has since received extensive observational and experimental support (Connell, 1961; MacArthur, 1972). The current theory holds that a species' response to spatial or temporal variation in temperature will depend both on direct effects on the individual- and population-level attributes of that species and on indirect effects mediated by changes in the distribution, abundance, and behavior of competitors, predators, parasites, and mutualists (Dunson and Travis, 1991; Davis et al., 1998; Sanford, 1999; Hawkins et al., 2009; Johnson et al., in press; Wernberg et al., in press). Thus, although general patterns of change may be robust and predictable (e.g. Barry et al., 1995; Parmesan and Yohe, 2003), accurate predictions regarding the consequences of warming for particular species or ecosystems of interest often remain elusive.

A significant challenge in this era of global change is to improve our predictive power with regards to the ecologically important consequences of climatic warming. To accomplish this, we must integrate single species, ecophysiological/population-level approaches and multispecies, community- and ecosystem-level research into a single framework so that general hypotheses regarding the effects of warming

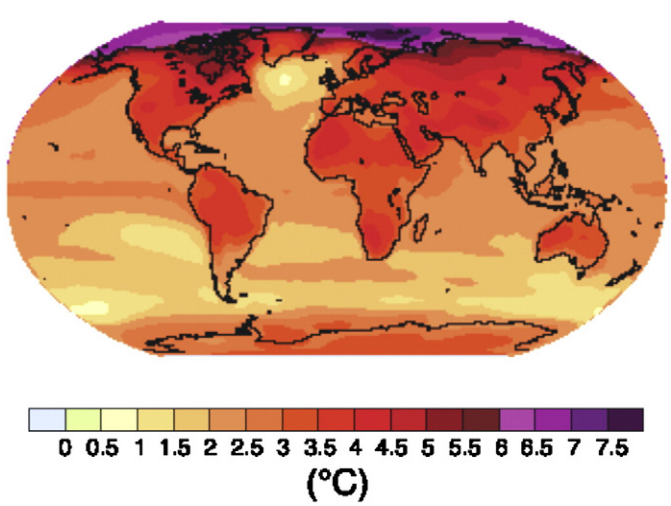

Fig. 1. Projected surface temperature changes for the late 21st century relative to the period 1980–1999. The panels show the multi-AOGCM average projections for the A1B SRES scenarios averaged over 2090–2099 (IPCC, 2007).

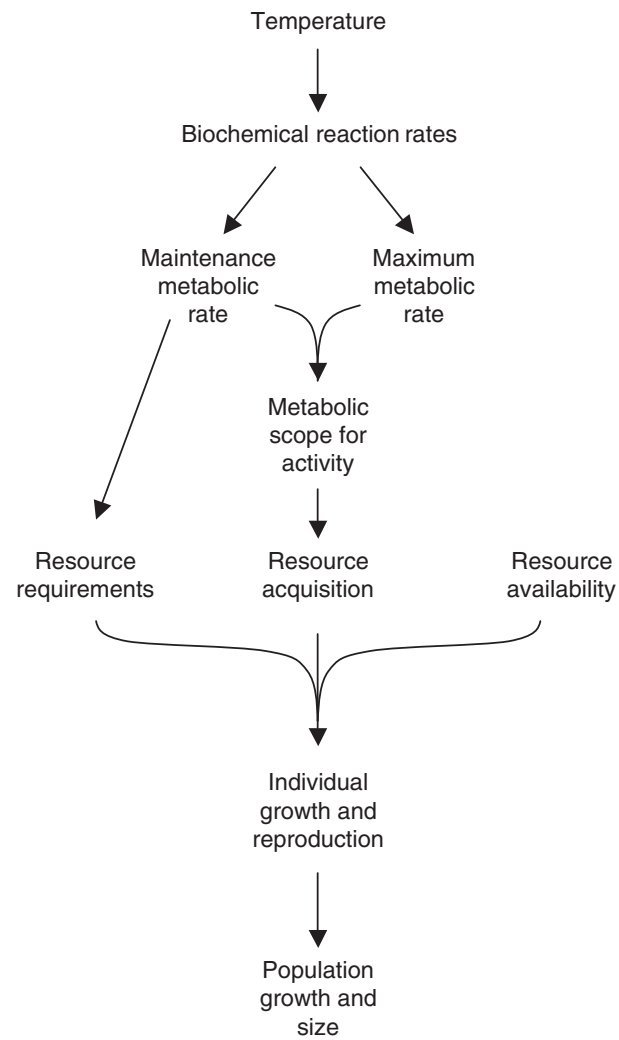

Fig. 2. The pathway by which temperature as a physical phenomenon influences the ecology of individuals and populations.

can be formulated and tested, and a theory of climate change ecology can progress. Here, we consider biological effects of temperature change across levels of organization from enzymes to ecosystems to determine how much is known about the potential effects of temperature on complex groups of interacting species. We begin with a brief review of how temperature affects basic metabolic processes, and then explore how differences in these responses among species affect species interactions. Next, we consider how differences in physiological responses across different species can influence the overall effect of temperature on ecological communities. Finally, we outline possible frameworks for generalization of the impacts of temperature on ecological systems, and consider broader implications of these generalities for climate change and biogeographic patters in marine systems. We do not intend to present an exhaustive review of the ever-expanding literature on climate change. Rather, we aim to highlight the ways in which warming will influence species interactions, and the ways in which species interactions will determine the outcome of warming.

2. The biological importance of temperature

Temperature is one of the most important factors affecting biological processes in poikilotherms (see Fig. 2 for a summary). The link between temperature and biological processes is kinetic; as temperature rises and atoms become more energetic, processes such as diffusion speed up and molecules in a fluid collide with one another

more frequently. For enzyme-catalyzed reactions, higher temperatures increase the likelihood that enzymes will collide and bind with substrate molecules during a given time frame, enhancing the speed and efficiency of biochemical reactions. However, enzymes are proteins that are largely held together by hydrogen bonds, and temperatures that exceed some threshold can weaken these bonds, causing proteins to change shape and thus reducing or negating their effectiveness as biological catalysts. Because enzymes work best within a specific temperature range, and because diffusion increases with temperature, catalytic rates typically increase with temperature to a point after which they fall off rapidly (Campbell and Farrell, 2006) (Fig. 3a).

Enzymatic reactions underlie functions at higher levels of organization; therefore, other biological rates often exhibit similar relationships with temperature. For example, metabolic function is strongly temperature dependent. For ectotherms, rising temperature increases the rates of basal metabolic rate and the rate at which energy stores are depleted. Temperature also determines the maximum metabolic rate, which determines the limits of non-maintenance activities such as exercise (via the breakdown of energy stores) and growth and reproductive investment (via the build-up of somatic and gonadal tissue). The difference between the active metabolic rate (the maximum rate at which an organism *can* expend energy, e.g., during activity) and the resting metabolic rate (the rate at which an organism *must* expend energy to stay alive and healthy, e.g., respiration) can be thought of as the metabolic scope for work. In essence, the metabolic scope for work is a proxy for the energy available for non-maintenance functions such as physical activity, growth, and reproduction (metabolic scope for work is therefore a broader term than the more commonly used 'metabolic scope for activity'; e.g. Claireaux and Lefrancois, 2007). As with biochemical reactions, scope for work increases from low temperature towards some optimum, and then begins to fall off as costs begin to accrue more rapidly than benefits (Lee et al., 2003) (Fig. 3b).

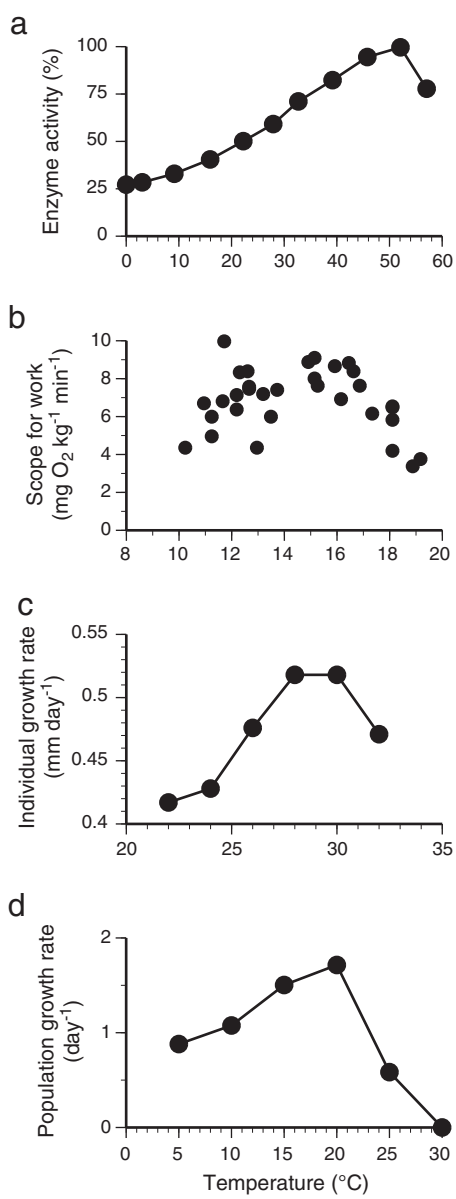

Fig. 3. Relationship between temperature and various biological rates for representative species (note the differences in x-axis scale). a. Activity of the enzyme lactate dehydrogenase in the fish *Champsocephalus gunnari* (Coquelle et al., 2007). b. Metabolic scope for work (measured as maximal metabolic rate minus resting metabolic rate) in sockeye salmon *Oncorhynchus nerka* (Lee et al., 2003). c. Individual growth rate in the Cortez oyster *Crassostrea corteziensis* (Caceres-Puig et al., 2007). d. Population growth rate of the marine diatom *Phaeodactylum tricornutum* (Kudo et al., 2000), using data for iron-replete cultures.

Because metabolic scope for activity represents energy available for non-maintenance functions, it is not surprising that individual growth rates display a similar unimodal relationship with temperature (Fig. 3c). Note that the temperature–growth relationship depends on food and other resources being amply supplied; if food is scarce, an organism may not meet its maintenance metabolic costs even though it is capable of high levels of activity. We will return to this idea when we discuss the Metabolic Theory of Ecology. Faster individual growth rates in turn tend to reduce generation time, and thermal control of generation time has important consequences for rates of population growth (Huey and Berrigan, 2001). Indeed, population growth rates frequently exhibit the same relationship with temperature as individual growth rates (Fig. 3d).

3. Interspecific variation in thermal sensitivity

Every species will exhibit some relationship between temperature and fundamental biological performance parameters such as metabolic rate and growth. However, the relationship between temperature and performance can vary widely among species. There are two fundamental ways in which this interspecific variation can manifest: 1) differences in thermal sensitivity (i.e., the slope of the temperature: performance relationship), and 2) differences in the maximum, minimum, or optimal temperatures for a given biological function. Variation in thermal sensitivity is diagrammed in Fig. 4a. In our hypothetical example, one species exhibits a relatively large increase in performance from low to optimal temperatures (Fig. 4a, dashed line), while another has a much more gradual increase in performance over the same range (Fig. 4a, solid line). Although both species have the same thermal range, the latter species (solid line) is less sensitive to changes in temperature, and will outperform the more thermally sensitive species at colder temperatures (point x) but not at warmer temperatures (point y). Alternatively, interacting species can have a difference in the position of the peak of their performance–temperature curve and in their thermal limits (Fig. 4b). In this case, the species represented by the solid line outperforms the other species at low temperature (point x), but is lost from the system at higher temperatures (point y).

Interspecific variation in thermal sensitivity is a general phenomenon. Before we present illustrative examples, however, we need a metric to describe the relationship between temperature and performance so that we may more easily compare thermal sensitivity among species. One such metric is the Q_{10} value, which is the factor by which performance (e.g., enzymatic reactions, metabolic rate, growth) increases with a 10 °C increase in temperature. For example, a Q_{10} of 2 implies a doubling of metabolic rate when temperature is increased from 10 °C to 20 °C (For drawbacks to using Q10, see

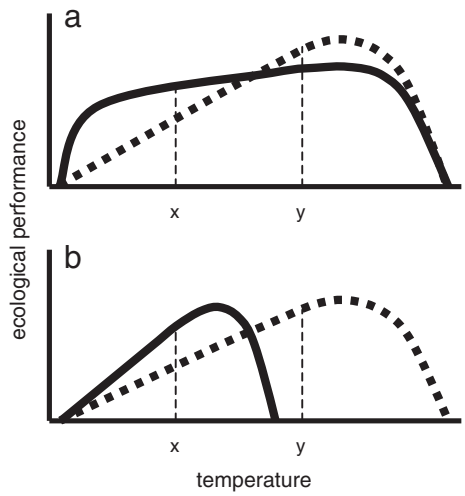

Fig. 4. Interspecific variation in the impact of rising temperatures. In the upper panel, the species represented by the dashed line is more sensitive to changes in the thermal environment across most temperatures, but each species has the same thermal range. In the lower panel, the dashed-line species has a higher upper thermal limit. If 'ecological performance' were to represent, e.g., competitive ability, an increase in temperature from x to y would result in a shift in competitive dominance from the solid-line species to the dashed-line species.

Gillooly et al., 2002). When Q_{10} values are compared among interacting taxa, they may vary considerably; Q_{10} values for northern European bivalve metabolic rates are near 2.0, while the Q_{10} values for the metabolic rates of species which prey on those bivalves can range from 1.5 to 2.5 (Freitas et al., 2007). When two species with different thermal sensitivities are allowed to interact, the outcome of that interaction is also temperature sensitive. For example, predatory flagellates are more sensitive to (i.e., respond more positively to) increases in temperature than do their bacterial prey (Delaney, 2003). Although Delaney (2003) was primarily concerned with the effects of turbulence, we can calculate approximate Q_{10} values from the data presented in her Tables 1 and 2 (using the turbulent treatment, which was considered a better approximation of natural conditions). The Q_{10} for the population growth rate of the predator (~3.4) was higher than that of the prey (~2.4), which would correspond to the dashed and solid lines in Fig. 4a, respectively. As a result of both relatively more rapid predator population increases and higher per capita predator ingestion rates at higher temperatures, the overall mortality of bacteria due to flagellate grazing increased over 5-fold for every 10 °C of warming (Delaney, 2003). Rising temperatures could therefore favor bacterial population growth in the absence of a predator but hinder bacterial population growth in the presence of a predator.

Not surprisingly, variation in thermal range or thermal optima among species within a community is also a widespread phenomenon that has important ecological consequences. For example, on New England rocky shores, two competing species of barnacles have different maximum temperature tolerances, and a combination of temperature and interspecific competition determines the distribution of the two species. In cooler, northern areas, thermally intolerant *Semibalanus balanoides* (represented by the solid line in Fig. 4b) competitively excludes the more thermally tolerant *Chthamalus fragilis* in the mid and high intertidal zones (dashed line, Fig. 4b). In warmer southern areas, high temperatures exclude *S. balanoides* from the higher shore levels and *C. fragilis* occupies that free space (Wethey, 1983, 1984). Similar relationships occur on European rocky shores with *S. balanoides* outcompeting *Chthamalus* species, in most cases (Connell, 1961). The importance of climatic fluctuations in mediating interactions between *S. balanoides* and *Chthamalus* species, have been long known (Southward and Crisp, 1954; Southward, 1991). Recent analysis of 40 year data sets and modeling (Poloczanska et al., 2008) have shown, in warmer years, *Chthamalus* species are released from competition with faster growing, cold-water *S. balanoides* in warm years.

As illustrated by the above examples, warming temperatures can affect a species via both direct and indirect pathways. There has been a great deal of emphasis on the direct impacts of temperature on ecological variables including local abundance. However, indirect effects such as the increase in *C. fragilis* observed when high temperature inhibits the dominant competitor may also be just as important (Poloczanska et al., 2008). These indirect effects can be divided into two categories: per capita effects, where temperature changes the strength of a single individual's interaction within a community, and density effects, where temperature changes in the total number of individuals in the population. Both mechanisms can and probably do operate simultaneously. For example, during periods of upwelling, when sea surface temperatures decrease, the sea star *Pisaster ochraceus* (Fig. 5a) becomes less abundant in the intertidal zone where it forages due to reduced activity (a population-level effect) (Fig. 5b). In addition, individual *Pisaster* consumes fewer mussels per unit time in colder water (a per capita effect) (Fig. 5c). The net effect of colder water is a dramatic decrease in the rate of mussel mortality due to predation (Sanford, 1999). Although mussels do grow more slowly in cooler water (Menge et al., 2008), a cooling-induced decrease in predation may more than offset this direct negative effect on mussel populations.

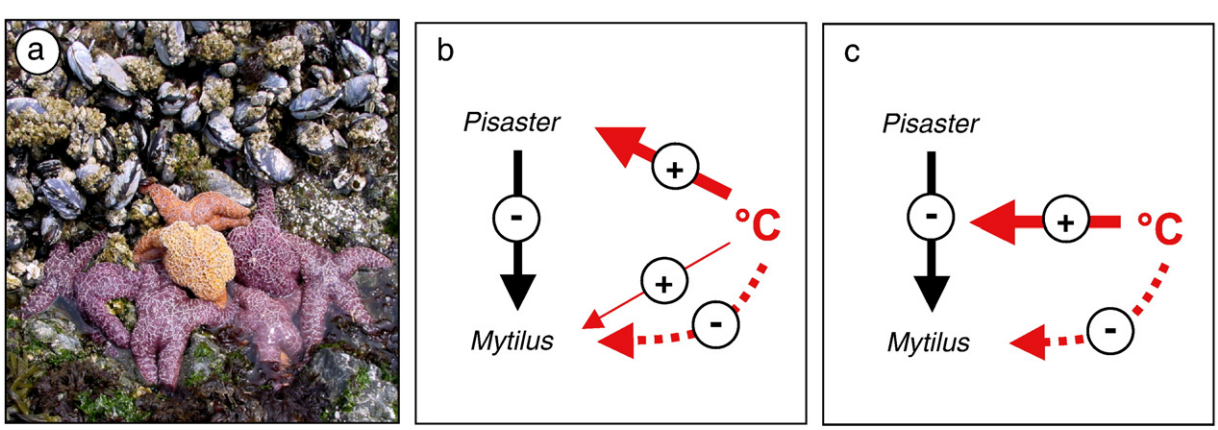

Fig. 5. Direct and indirect effects of rising temperature (T) on an interacting species pair. a) *Pisaster ochraceus* and *Mytilus californianus*. b) Density effects, where rising temperature increases *Pisaster* abundance (solid, thick red arrow). c) Per capita effects, where the strength of predation (black arrow) is increased (made more negative) by rising temperatures (solid, thick red arrow). In both the per capita and density-mediated cases, the net effect of temperature on mussels is negative (dashed red line) despite any weak direct effects to the contrary (thin red line in panel b).

Indirect effects mediated by species such as *Pisaster* may determine much of the net effect of warming at the community and ecosystem levels. As noted by Sanford (1999), key species interactions that are sensitive to temperature may act as "leverage points" through which small changes in climate could generate large changes in natural communities. Species that act on these leverage points can amplify the signal of small changes in climate to generate unexpectedly large changes at the community level. In addition to classic keystone species such as *Pisaster*, many diseases and pests are likely to operate on leverage points. For example, warming increases the incidence and impact of pathogens in many marine species (Harvell et al., 2002), including *Pisaster* (Bates et al., 2009). This further highlights some of the potential complexities involved; *Pisaster* predation may increase with temperature, but over the longer term this effect may depend on the presence and epidemiology of sea star disease agents.

4. Incorporating time and space: phenology and biogeography

A change in temperature can alter species interactions if the sign or magnitude of response differs among the species (Fig. 4). Species interactions may also change if temperature causes a change in the temporal or spatial abundance pattern of one of the species relative to another. Climatic warming is causing spring to start earlier and summer to last longer (Menzel and Fabian, 1999; Thompson and Clark, 2008), and as a result many plant and animal phenologies (the timing of reproduction, larval release or settlement, fledging, migration, etc.) are also shifting earlier (Sims et al., 2001; Philippart et al., 2003; Edwards and Richardson, 2004; Hays et al., 2005). Parmesan and Yohe (2003) showed that over 45 (median) years 62% of 678 species worldwide have exhibited changed phenologies. In addition, a meta-analysis of 203 species spanning the northern hemisphere revealed an advance in spring-cued phenology of 2.8 days/decade (Parmesan, 2007), and coastal marine species are moving even faster (Helmuth et al., 2006b). Moore et al. (2011) have recently showed that whilst a southern species of limpet (*P. depressa*) is breeding earlier and longer, a northern autumn breeding congener is breeding later and failing to breed in some years.

The timing of life cycle transitions must often be in (or out of) synchrony with the phenology of other species, particularly when those species represent an important food resource or an important source of mortality. For example, the timing of hatching or spawning often occurs when food resources will be most plentiful for offspring (e.g. Platt et al., 2003). Mismatches between periods of larval presence and planktonic food abundance associated with interannual climate variability have been long been blamed for poor fisheries yields (Cushing, 1982). Although consumers can be cued by their resource directly, many must rely on some perceptible environmental cue such as temperature or light as a proxy for it. Although most documented cases of this phenomenon have been from terrestrial systems, recent work in marine systems, primarily on seabirds and pelagic communities, have highlighted how linked species can be cued by different factors (Costello et al., 2006; Richardson, 2008; Watanuki et al., 2009). For example, (Edwards and Richardson, 2004) analyzed data from 66 marine taxa spanning more than 40 years and found that diatom blooms have remained fixed in time (cued by light) while temperature-cued consumers have shifted reproduction earlier as summer water temperatures increase. This has led to a phenological mismatch between trophic levels.

Differential use of the thermal landscape can also lead to temporal mismatches. Thermal cues in migratory animals' wintering grounds are becoming less predictive of conditions on the breeding grounds. Indeed, many migrant animals rely on a series of locations during the year, each with a different climatic regime, each changing at a different rate with global warming (i.e., Jonsson and Jonsson, 2009). Historically, migrant animals have arrived at their breeding grounds in synchrony with their food source. However some but not necessarily all sites along a migration route are being affected by warming, thus animals end up mistimed with their resource at their reproductive locations (Carscadden et al., 1997; Sims et al., 2004).

As formerly relevant seasonal cues lose their accuracy in matching resources and environmental conditions, phenological mismatches are becoming common. One study reviewed cases where species had become mistimed to see if they had fallen too far out of alignment, and found that out of 11 cases, eight had become uncoupled, shifting either too soon or too late compared to the other (Visser and Both, 2005). The most pertinent question may be whether these mismatched species remain uncoupled, or whether ecological or evolutionary processes can compensate for negative consequences of the mismatch. For example, selection or plasticity in phenology could act strongly enough to re-couple them over time, or to facilitate prey switching or other behavioral shifts to compensate for climate impacts.

Biogeographic range shifts are another obvious biological manifestation of climatic warming (e.g. Helmuth et al., 2006b); 75% of 129 coastal marine species have undergone poleward shifts in their geographic distributions, at an average rate of 19 km/year (Sorte et al., 2010). Warming-induced range shifts may widely alter the compliment of interacting species at a site (Cheung et al., 2009), and interspecific interactions may determine the extent to which any given species range changes with warming. We consider each of these scenarios in turn.

Biogeographic range shifts during times of environmental change are nothing new in the earth's history, and the fossil record can shed a great deal of light on the implications of ongoing and future range shifts. Analyses of post ice age warming in the late-Quaternary indicates that some species shift their range limits during periods of warming while others do not (Roy et al., 2001). Such individualistic responses among species can cause historically separated species ranges to converge, potentially generating a new interspecific interaction, or force interacting species apart geographically and eliminate an interspecific interaction (Fig. 6). This reshuffling of taxa results in combinations of species that cannot be found together anywhere on earth at present—a situation known as a no-analog community (Williams and Jackson, 2007). There are several marine examples of no-analog communities during the recent geological past when global temperatures differed considerably from the present (e.g., Kitamura, 2004; Steinke et al., 2008), and more no-analog communities can be expected in the future. One critical question is whether no-analog communities differ in their structure or functioning relative to communities in which evolution may have led to sets of

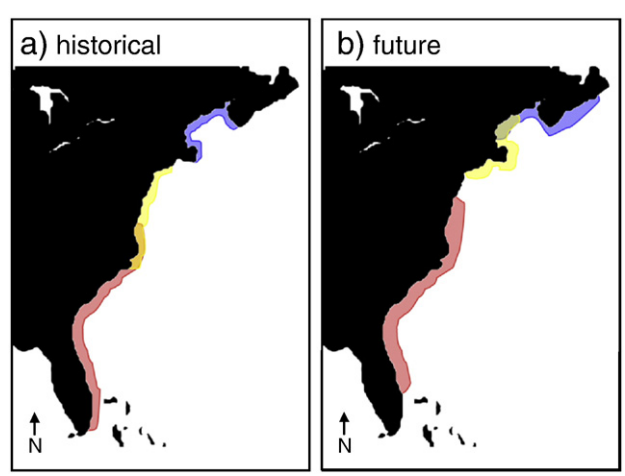

Fig. 6. Hypothetical range shifts due to global warming with the resulting species interactions. a) The yellow species' historical range overlaps with that of the red species. b) Warming may cause species' ranges to move poleward, but to different extents, generating novel interactions, such as with the yellow and blue species.

traits that allow greater function or unique community structure. Many no-analog communities already exist as a consequence of human mediated species introductions. While there is substantial evidence that biological invasions can change community structure through cascades of interactions (Grosholz et al., 2000; Wonham et al., 2005), there is not clear evidence that novel combinations of species within a community consistently alter community structure or functioning relative to uninvaded communities, though the role of temperature in this context has not been examined explicitly.

There is no doubt that biogeographic changes like those recorded in the fossil record are ongoing today. In recent decades, warming has triggered an expansion of species' poleward range boundaries and a contraction of equatorward range boundaries (Sagarin et al., 1999; Perry et al., 2005; Southward et al., 1995, 2005; Helmuth et al., 2006b; Moore et al., 2007a; Sorte et al., 2010). The velocity of these modern shifts can be striking; the southern range limit of a barnacle and the northern range limit of a benthic polychaete are moving north at rates of 15–50 km/decade in Europe (Wethey and Woodin, 2008), and some planktonic species are moving an order of magnitude faster than that (Beaugrand et al., 2002; Hays et al., 2005).While some species ranges are shifting quickly, others are shifting slowly, and still others are either not shifting at all or are moving in the opposite direction (e.g., Perry et al., 2005; Lima et al., 2007). As with no-analog communities of the past, this complex redistribution of species guarantees that some species will exchange encounters with familiar organisms for interactions with novel organisms. For example, global warming is facilitating the poleward spread of many harmful algal bloom species, creating risks for wildlife and human health in previously unimpacted areas in both hemispheres (Hallegraeff, 2010).

Changes analogous to these latitudinal shifts are also occurring across vertical gradients of depth and intertidal height. The depth distributions of North Sea fishes are generally shifting to deeper waters, although the degree and even direction of depth range change is species specific (Perry et al., 2005). Although the ecological implications of any resulting shifts in interspecific interactions remain largely unknown for fish assemblages, some data is available for redistributions of benthic species across the vertical gradient. On rocky shores in the northeast Pacific, rising temperatures have forced the upper limits of the alga *Mazzaella parksii* to lower positions on the shore (Harley and Paine, 2009). Although the upper limit of the alga is related directly to temperature via the species' environmental tolerance, the lower limit (set by molluscan grazers) is independent of temperature. Higher temperatures result in an increase of the spatial overlap between the potential vertical range of *Mazzaella* and that of its consumers, which in turn leads to the elimination of the alga in warm areas which lack a spatial refuge from herbivory (Harley, 2003).

This latter example is illustrative of the potential role of interspecific interactions in determining the degree to which species ranges may expand or contract with warming. To remain within its current thermal envelope, *Mazzaella* would have had to shift both its upper and lower limits downshore. However, consumers prevented such a shift in the lower limit, with negative implications for the total vertical range of the alga. Interactions among species are known to determine the position of range limits along a thermal gradient in the laboratory (Davis et al., 1998). The degree to which species interactions may generally facilitate or inhibit species' ability to track their preferred environmental conditions (often called it's bioclimatic envelope) in the field, particularly at larger spatial scales, remains an open question.

5. The search for generality

Although the effect of temperature on the performance of an individual, population or species varies from case to case, considering generalities of biological effects of temperature allows the articulation of testable hypotheses and exploration of potentially broad-scale impacts of temperature on communities and ecosystems. Recent work suggests that interspecific interactions shift from generally negative (e.g. competitive) when the environment is benign to generally positive (e.g., facilitative) when the environment is stressful (Bruno et al., 2003). Since high temperature can qualify as an environmental stress, many interactions are predicted to shift from competitive to facilitative at higher temperatures (Wernberg et al., 2010). This has been shown to occur within a community-type; for example, canopy-forming algae on rocky shores compete with barnacles for space at cool sites but facilitate them by providing cool understory micro-habitats at warm sites (Leonard, 2000). In the same ecosystem, the per capita effect of *S. balanoides* on fucoid germlings varies among environments (latitudinally) and between barnacle life stages (Kordas and Dudgeon, 2010). Facilitation theory is relevant to systems where species interactions can ameliorate physical or physiological stress, and intertidal rocky shores or marshes are emblematic habitat types for facilitation. It is less clear how the prevalence, strength or importance of facilitation will change in subtidal communities where organisms cannot modify the temperature of the ocean. This difference in facilitation across habitat types is supported by a comparison among community types (e.g. warmer high-shore barnacle dominated communities vs. cooler low-shore kelp-dominated communities) in which the relative importance of competitive and facilitative interactions does not appear to change (Wood et al., 2010). Facilitation may also favour one species more than others: Moore et al. (2007a) showed that the behaviour of the Northern species of limpet *Patella vulgata* allowed it to benefit from habitual amelioration by fucoid clumps; whilst its more southerly congener, *P. depressa* did not display such behaviour. These changes have implications for patch dynamics and functioning of European rocky shores (Hawkins et al., 2008, 2009).

The broader search for generalities in ecology has led to the development of the Metabolic Theory of Ecology (MTE), which relates metabolic rate to body size and temperature (Gillooly et al., 2001). MTE predicts that metabolic rate increases with temperature in specific ways across broad taxonomic groups (Gillooly et al., 2001). However, at this coarse resolution, differences among some groups persist. Exploring these differences at the group level (e.g., primary- versus secondary-producers, fish versus invertebrates (Gillooly et al., 2001; López-Urrutia et al., 2006)) may lead to general patterns in how community structure (relative abundance of species or functional groups) varies with temperature change. In this way, a theory of how temperature affects community structure can be developed and tested.

Specific physiological rates may also respond differently to changes in temperature. For example, both theoretical and empirical evidence suggests that marine planktonic respiration increases more rapidly with rising temperature than does photosynthesis (López-Urrutia et al., 2006). Thus, rising temperatures should shift marine planktonic systems away from autotrophy and towards heterotrophy, a prediction which has some empirical support (Müren et al., 2005).

MTE also makes predictions regarding the relationship between temperature and food web structure. Food chain length depends on the amount of energy transferred through trophic interactions. For a given (fixed) resource base, the highest trophic level in the system is that which can support a minimum viable population with the energy available from lower trophic levels. As temperature increases, the metabolic rates of all species increase, resulting in an increasing demand for and consumption of energy at each trophic level. When the supply of energy transferred up the food chain is no longer sufficient to support the minimum viable population size of the top predator, that species is lost. There is some empirical support for this prediction; by experimentally warming mesocosms containing aquatic microbes, Petchey et al. (1999) found that higher trophic levels were lost disproportionately, and food chain length decreased. As consumers were lost in warmed treatments, primary producer and bacterivore biomass increased; suggesting that a thermally-triggered

trophic cascade had occurred (Petchey et al., 1999). Although these results are consistent with MTE predictions, other alternatives, such as lower physiological tolerance to warming in species at higher trophic levels, cannot be ruled out. Furthermore, in surface and coastal marine systems, ocean currents and nutrient availability change with temperature in varied ways. Changes in upwelling will increase nutrient availability while constraining temperature changes, while in other areas increased thermal stratification will reduce nutrients concurrent with warming. Changes in upwelling will also influence recruitment regimes (e.g. Menge et al., in press). Any general effects of temperature on species interactions will occur in the context of other, potentially more influential environmental changes. The principal challenge at this stage is to develop and test predictions for how these changes interact to influence species interactions to determine whether any generalities exist.

MTE has been useful for generating broad-scale models of the ecological responses to temperature change. It is less clear whether MTE applies to smaller spatial and temporal scales, where species' traits and differences may be more important. In this case, more detailed theories like dynamic energy budgets may be more relevant for generating predictions (Helmuth et al., 2006a).

6. Future research priorities

Anthropogenic climate change is creating an ongoing series of challenges for human societies that rely on natural goods and services. At present, our lack of understanding of the interplay between temperature and interspecific interactions prevents ecologists from making anything more than relatively basic predictions regarding the effects warming on community structure, on ecosystem function, and even on individual species of concern. The degree to which future outcomes will follow predictable patterns based on general species attributes (e.g., trophic level) or will only be predictable with careful study of the individual species involved remains unclear. In either case, predictions for the future inherently require extrapolation beyond the current range of observations, and therefore require the application of basic, mechanistic ecological principles to new situations (e.g. Poloczanska et al., 2008). A stronger mechanistic understanding of climate change impacts can be achieved through a systematic approach that emphasizes the testing of hypotheses in experimental frameworks (Firth et al., 2009). This fundamental scientific method has not been emphasized in climate change ecology, in part because the focus has been on documentation of impacts. The current challenge is now to determine the extent to which we can understand the causes and consequences of these impacts in a general ecological framework. Currently, numerous hypotheses based on physiological, ecological and evolutionary theory can be articulated and experimentally tested. We outline a few key questions here:

- Can among-species variation in thermal sensitivity (i.e., the slope of the temperature: performance relationship) or critical temperatures (thermal optima, maximum or minimum temperatures for a given biological function) predict how interactions such as competition and predation will change with warming? Some evidence suggests that this approach may bear fruit, particularly for trophic relationships where production and consumption rates can be carefully measured (Delaney, 2003). However, at least one classic competition example (Park, 1954) shows that surpassing the growth rate of a superior competitor at higher temperature does not lead to a switch in competitive dominance, and simple comparisons of growth rates may be misleading in light of the potential trade-off between growth rate and competitive ability.
- To what extent will ecological change be driven by changes in abundance of interacting species (population-level effects) vs. changes in per capita effects? Although much of the ecological literature focuses on the relative change in abundance of strong interactors (e.g., predators, ecosystem engineers, disease vectors), which is easier to measure, the flour beetle example mentioned above (Park, 1954) along with more recent work (e.g. Sanford, 1999; Moore et al., 2007b) suggests that per capita interactions may be critical. Integrating these two levels of impact is a priority because they can driven by different mechanisms and therefore may change at different rates with environmental change, and be subject to different constraints and limitations.
- Is the shift from predominantly negative interactions to predominantly positive interactions as stress increases—a phenomenon which holds for specific, defined assemblages (e.g. Leonard, 2000) in habitats where organisms can modify the thermal environment—likely to apply when species composition is also changing? What is the role of facilitation in subtidal systems where organisms are not able to modify their thermal environment?
- How much can the Metabolic Theory of Ecology tell us about specific communities? Is the loss of top predators during periods of warming a general phenomenon? And, as with the question of positive versus negative interactions, does the decrease in food chain length only apply when novel, thermally tolerant species are not allowed to invade the system?
- To what extent will evolution minimize or even exacerbate community-level responses to warming? Local adaptation to the thermal environment is well documented, and mechanistic predictions developed using present-day thermal tolerance limits, temperature–performance functions, or phenological relationships to temperature may not apply in the future.

Answers to these questions will require studies that simultaneously address physiological responses to abiotic variables and ecological relationships among interacting species. Dunson and Travis (1991) lamented the scarcity of such studies two decades ago, and there is still a great need to unify ecophysiology and community ecology. Such research will be necessary to field-test hypotheses that have been developed on the basis of thermodynamic considerations and laboratory results. Studying ecological dynamics in artificially warmed areas such as power plant cooling water discharge plumes (e.g. Schiel et al., 2004) or during warm phases of natural climatic cycles (e.g. ENSO) is a good start, but well-designed thermal manipulations (e.g. Harte and Shaw, 1995; McKee et al., 2003) that test responses of critical ecological and evolutionary processes in the context of theory are badly needed. Furthermore, although much can be learned from the paleo-ecological perspective, ongoing research must incorporate potential synergisms between warming and other modern anthropogenic effects such as habitat modification, species introductions, over-exploitation, pollution, and elevated carbon dioxide (Williams and Jackson, 2007). Finally, our current predictions of future change are founded on present-day physiological and ecological responses to temperature, but organisms can acclimate and species can evolve. Although there has not yet been any evidence of genetic changes in populations towards higher thermal tolerances, populations can track climatic shifts to varying degrees through genetic change or plasticity (Bradshaw and Holzapfel, 2006). The extent to which phenotypic plasticity and natural selection will offset the effects of warming (e.g., shifts towards thermally tolerant genotypes of coral endosymbionts (Jones et al., 2008)) is poorly understood at best. Although the challenges are many, a fuller understanding of the complexities surrounding community-level responses to warming is a prerequisite for successfully predicting, mitigating, and managing the effects of global warming.

Acknowledgements

We thank Stefan Storey, Tony Farrell, Eric Sanford, Trish Schulte, and David Inouye for constructive criticisms on earlier drafts of this manuscript. **[SS]**

References

Barry, J., Baxter, C., Sagarin, R., Gilman, S., 1995. Climate-related, long-term faunal changes in a California rocky intertidal community. Science 267, 672–675.

Bates, A.E., Hilton, B.J., Harley, C.D.G., 2009. Effects of temperature, season and locality on wasting disease in the keystone predatory sea star *Pisaster ochraceus*. Dis. Aquat. Organ. 86, 245–251.

Beaugrand, G., Reid, P.C., Ibanez, F., Lindley, J.A., Edwards, M., 2002. Reorganization of North Atlantic marine copepod biodiversity and climate. Science 296, 1692–1694.

Bradshaw, W.E., Holzapfel, C.M., 2006. Climate change—evolutionary response to rapid climate change. Science 312, 1477–1478.

Bruno, J.F., Stachowicz, J.J., Bertness, M.D., 2003. Inclusion of facilitation into ecological theory. Trends Ecol. Evol. 18, 119–125.

Caceres-Puig, C.E., Abasolo-Pacheco, F., Mazon-Sastegui, J.M., Maeda-Martinez, A.N., Saucedo, P.E., 2007. Effect of temperature on growth and survival of *Crassostrea corteziensis* spat during late-nursery culturing at the hatchery. Aquaculture 272, 417–422.

Campbell, M.K., Farrell, S.O., 2006. Biochemistry, Fifth edition. Thomson Brooks/Cole, Belmont, CA.

Carscadden, J., Nakashima, B.S., Frank, K.T., 1997. Effects of fish length and temperature on the timing of peak spawning in capelin (*Mallotus villosus*). Can. J. Fish. Aquat. Sci. 54, 781–787.

Cheung, W.W.L., Lam, V.W.Y., Sarmiento, J.L., Kearney, K., Watson, R., Pauly, D., 2009. Projecting global marine biodiversity impacts under climate change scenarios. Fish Fish. 10, 235–251.

Claireaux, G., Lefrancois, C., 2007. Linking environmental variability and fish performance: integration through the concept of scope for activity. Philos. Trans. R. Soc. Lond. B Biol. Sci. 362, 2031–2041.

Connell, J.H., 1961. The influence of interspecific competition and other factors on the distribution of the barnacle *Chthamalus stellatus*. Ecology 42, 710–723.

Coquelle, N., Fioravanti, E., Weik, M., Vellieux, F., Madern, D., 2007. Activity, stability and structural studies of lactate dehydrogenases adapted to extreme thermal environments. J. Mol. Biol. 374, 547–562.

Costello, J., Sullivan, B., Gifford, D., 2006. A physical–biological interaction underlying variable phenological responses to climate change by coastal zooplankton. J. Plankton Res. 28, 1099–1105.

Cushing, D.H., 1982. Climate and Fisheries. Academic Press, London.

Darwin, C.R., 1959. On the Origin of Species. John Murray, London.

Davis, A.J., Jenkinson, L.S., Lawton, J.H., Shorrocks, B., Wood, S., 1998. Making mistakes when predicting shifts in species range in response to global warming. Nature 391, 783–786.

Delaney, M., 2003. Effects of temperature and turbulence on the predator–prey interactions between a heterotrophic flagellate and a marine bacterium. Microb. Ecol. 45, 218–225.

Dunson, W.A., Travis, J., 1991. The role of abiotic factors in community organization. Am. Nat. 138, 1067–1091.

Edwards, M., Richardson, A.J., 2004. Impact of climate change on marine pelagic phenology and trophic mismatch. Nature 430, 881–884.

Firth, L.B., Crowe, T.P., Moore, P., Thompson, R.C., Hawkins, S.J., 2009. Predicting impacts of climate-induced range expansion: an experimental framework and a test involving key grazers on temperate rocky shores. Gl. Ch. Biol. 15, 1413–1422.

Freitas, V., Campos, J., Fonds, M., Vanderveer, H., 2007. Potential impact of temperature change on epibenthic predator–bivalve prey interactions in temperate estuaries. J. Therm. Biol 32, 328–340.

Gillooly, J.F., Brown, J.H., West, G.B., Savage, V.M., Charnov, E.L., 2001. Effects of size and temperature on metabolic rate. Science 293, 2248–2251.

Gillooly, J.F., Charnov, E.L., West, G.B., Savage, V.M., Brown, J.H., 2002. Effects of size and temperature on developmental time. Nature 417, 70–73.

Grosholz, E.D., Ruiz, G.M., Dean, C.A., Shirley, K.A., Maron, J.L., Connors, P.G., 2000. The impacts of a nonindigenous marine predator in a California bay. Ecology 81, 1206–1224.

Hallegraeff, G.M., 2010. Ocean climate change, phytoplankton community responses, and harmful algal blooms: a formidable predictive challenge. J. Phycol. 46, 220–235.

Harley, C.D.G., 2003. Abiotic stress and herbivory interact to set range limits across a two-dimensional stress gradient. Ecology 84, 1477–1488.

Harley, C.D.G., Paine, R.T., 2009. Contingencies and compounded rare perturbations dictate sudden distributional shifts during periods of gradual climate change. Proc. Natl. Acad. Sci. USA 106, 11172–11176.

Harte, J., Shaw, R., 1995. Shifting dominance within a montane vegetation community: results of a climate-warming experiment. Science 267, 876–880.

Harvell, C.D., Mitchell, C.E., Ward, J.R., Altizer, S., Dobson, A.P., Ostfeld, R.S., Samuel, M.D., 2002. Climate warming and disease risks for terrestrial and marine biota. Science 296, 2158–2162.

Hawkins, S.J., Moore, P., Burrows, M.T., Poloczanska, E., Mieszkowska, N., Jenkins, S.R., Thompson, R.C., Genner, M.J., Southward, A.J., 2008. Complex interactions in a rapidly changing world: responses of rocky shore communities to recent climate change. Clim. Res. 37, 123–133.

Hawkins, S.J., Sugden, H.E., Mieszkowska, N., Moore, P., Polczanska, E., Leaper, R., Herbert, R.J.H., Genner, M.J., Moschella, P.S., Thompson, R.C., Jenkins, S.R., Southward, A.J., Burrows, M.T., 2009. Consequences of climate driven biodiversity changes for ecosystem functioning of North European Rocky Shores. Mar. Ecol. Prog. Ser. 396, 245–259.

Hays, G.C., Richardson, A.J., Robinson, C., 2005. Climate change and marine plankton. Trends Ecol. Evol. 20, 337–344.

Helmuth, B., 2009. From cells to coastlines: how can we use physiology to forecast the impacts of climate change? J. Exp. Biol. 212, 753–760.

Helmuth, B., Broitman, B.R., Blanchette, C.A., Gilman, S., Halpin, P., Harley, C.D.G., O'Donnell, M.J., Hofmann, G.E., Menge, B., Strickland, D., 2006a. Mosaic patterns of thermal stress in the rocky intertidal zone: implications for climate change. Ecol. Monogr. 76, 461–479.

Helmuth, B., Mieszkowska, N., Moore, P., Hawkins, S.J., 2006b. Living on the edge of two changing worlds: forecasting the responses of rocky intertidal ecosystems to climate change. Ann. Rev. Ecol. Evol. Sys. 37, 373–404.

Hochachka, P.W., Somero, G.N., 2002. Biochemical Adaptation: Mechanism and Process in Physiological Evolution. Oxford University Press, New York.

Huey, R.B., Berrigan, D., 2001. Temperature, demography, and ectotherm fitness. Am. Nat. 158, 204–210.

Hutchins, L.W., 1947. The bases for temperature zonation in geographical distribution. Ecol. Monogr. 17, 325–335.

IPCC, 2007. Climate Change 2007: Synthesis Report, p. 52.

Johnson, C.R., Banks, S.C., Barrett, N.S., Cazassus, F., Dunstan, P.K., Edgar, G, J., Frusher, S.D., Gardner, C., Haddon, M., Helidoniotis, F., Hill, K.L., Holbrook, N.J., Hosie, G.W., Last, P.R., Ling, S.D., Melbourne-Thomas, J., Swadling, K.M., Taw, N., in press. Climate change cascades: shifts in oceonagraphy, species' ranges and subtidal marine community dynamics in eastern Tasmania. J. Exp. Mar. Biol. Ecol.

Jones, A.M., Berkelmans, R., Mieog, J.C., Van Oppen, M.J.H., Sinclair, W., 2008. A community change in the symbionts of a scleractinian coral following a natural bleaching event: field evidence of acclimatization. Proc. R. Soc. B. 275, 1359–1365.

Jonsson, B., Jonsson, N., 2009. A review of the likely effects of climate change on anadromous Atlantic salmon *Salmo salar* and brown trout *Salmo trutta*, with particular reference to water temperature and flow. J. Fish Biol. 75, 2381–2447.

Kitamura, A., 2004. Effects of seasonality, forced by orbital-insolation cycles, on offshore molluscan faunal change during rapid warming in the Sea of Japan. Palaeogeogr. Palaeoclim. Palaoecol. 203, 169–178.

Kordas, R.L., Dudgeon, S., 2010. Dynamics of species interaction strength in space, time and with developmental stage. Proc. R. Soc. B. doi:10.1098/rspb.2010.2246.

Kudo, I., Miyamoto, M., Noiri, Y., Maita, Y., 2000. Combined effects of temperature and iron on the growth and physiology of the marine diatom, *Phaeodactylum tricornutum* (Bacillariophyceae). J. Phyc. 36, 1096–1102.

Lee, C.G., Farrell, A.P., Lotto, A., MacNutt, M.J., Hinch, S.G., Healey, M.C., 2003. The effect of temperature on swimming performance and oxygen consumption in adult sockeye (*Oncorhynchus nerka*) and coho (*O. kisutch*) salmon stocks. J. Exp. Biol. 206, 3239–3251.

Leonard, G.H., 2000. Latitudinal variation in species interactions: a test in the New England rocky intertidal zone. Ecology 81, 1015–1030.

Lima, F.P., Ribeiro, P.A., Quieroz, N., Hawkins, S.J., Santos, A.M., 2007. Do distributional shifts of northern and southern species of algae match the warming pattern? Gl. Ch. Biol. 13, 2592–2604.

López-Urrutia, A., San Martin, E., Harris, R.P., Irigoien, X., 2006. Scaling the metabolic balance of the oceans. Proc. Natl. Acad. Sci. USA 103, 8739–8744.

MacArthur, R., 1972. Geographical Ecology. Princeton University Press, Princeton, NJ.

McKee, D., Atkinson, D., Collings, S.E., Eaton, J.W., Gill, A.B., Harvey, I., Hatton, K., Heyes, T., Wilson, D., Moss, B., 2003. Response of freshwater microcosm communities to nutrients, fish, and elevated temperature during winter and summer. Limn. Ocean. 48, 707–722.

Menge, B.A., Chan, F., Lubchenco, J., 2008. Response of a rocky intertidal ecosystem engineer and community dominant to climate change. Ecol. Lett. 11, 151–162.

Menge, B.A., Gouhier, T.C., Freidenburg, T., Lubchenco, J., in press. Linking long-term, large-scale climatic and environmental variability to patterns of marine invertebrate recruitment: toward explaining "unexplained" variation. J. Exp. Mar. Biol. Ecol.

Menzel, A., Fabian, P., 1999. Growing season extended in Europe. Nature 397, 659-659.

Mieszkowska, N., Hawkins, S.J., Burrows, M.T., Kendall, M.A., 2007. Long-term changes in the geographic distribution and population structures of *Osilinus lineatus* (Gastropoda: Trochidae) in Britian and Ireland. J. Mar. Biol. Assoc. UK 87, 537–545.

Moore, P., Hawkins, S.J., Thompson, R.C., 2007a. Role of biological habitat amelioration in altering the relative responses of congeneric species to climate change. Mar. Ecol. Prog. Ser. 334, 11–19.

Moore, P., Thompson, R.C., Hawkins, S.J., 2007b. Effect of grazer identity on the probability of escapes by a canopy-forming macroalga. J. Exp. Mar. Biol. Ecol. 344, 170–180.

Moore, P., Thompson, R.C., Hawkins, S.J., 2011. Phenological changes in intertidal conspecific gastropods in response to climate warming. Gl.Ch.Biol. 17, 709–719.

Müren, U., Berglund, J., Samuelsson, K., Andersson, A., 2005. Potential effects of elevated sea-water temperature on pelagic food webs. Hydrobiologia 545, 153–166.

Park, T., 1954. Experimental studies of interspecies competition II. Temperature, humidity, and competition in two species of *Tribolium*. Physiol. Zool. 28, 177–238.

Parmesan, C., 2007. Influences of species, latitudes and methodologies on estimates of phenological response to global warming. Gl. Ch. Biol. 13, 1860–1872.

Parmesan, C., Yohe, G., 2003. A globally coherent fingerprint of climate change impacts across natural systems. Nature 421, 37–42.

Perry, A.L., Low, P.J., Ellis, J.R., Reynolds, J.D., 2005. Climate change and distribution shifts in marine fishes. Science 308, 1912–1915.

Petchey, O.L., McPhearson, P.T., Casey, T.M., Morin, P.J., 1999. Environmental warming alters food-web structure and ecosystem function. Nature 402, 69–72.

Philippart, C.J.M., van Aken, H.M., Beukema, J.J., Bos, O.G., Cadee, G.C., Dekker, R., 2003. Climate-related changes in recruitment of the bivalve *Macoma balthica*. Limn. Ocean. 48, 2171–2185.

Platt, T., Fuentes-Yaco, C., Frank, K.T., 2003. Spring algal bloom and larval fish survival. Nature 423, 398–399.

Poloczanska, E.S., Hawkins, S.J., Southward, A.J., Burrows, M.T., 2008. Modelling the response of populations of competing species to climate change. Ecology 89 (11), 3138–3149.

Richardson, A.J., 2008. In hot water: zooplankton and climate change. ICES J. Mar. Sci. 65, 279–295.
Roy, K., Jablonski, D., Valentine, J.W., 2001. Climate change, species range limits and body size in marine bivalves. Ecol. Lett. 4, 366–370.
Sagarin, R.D., Barry, J.P., Gilman, S.E., Baxter, C.H., 1999. Climate-related change in an intertidal community over short and long time scales. Ecol. Monogr. 69, 465–490.
Sanford, E., 1999. Regulation of keystone predation by small changes in ocean temperature. Science 283, 2095–2097.
Schiel, D.R., Steinbeck, J.R., Foster, M.S., 2004. Ten years of induced ocean warming causes comprehensive changes in marine benthic communities. Ecology 85, 1833–1839.
Sims, D.W., Genner, M.J., Southward, A.J., Hawkins, S.J., 2001. Timing of squid migration reflects North Atlantic climate variability. Proc. R. Soc. B. 268, 2607–2611.
Sims, D.W., Wearmouth, V.J., Genner, M.J., Southward, A.J., Hawkins, S.J., 2004. Low-temperature-driven early spawning migration of a temperate marine fish. J. Anim. Ecol. 73, 333–341.
Sorte, C.J.B., Williams, S.L., Carlton, J.T., 2010. Marine range shifts and species introductions: comparative spread rates and community impacts. Gl. Ecol. Biogeogr. 19, 303–316.
Southward, A.J., 1991. Forty years of changes in species composition and population density of barnacles on a rocky shore near Plymouth. J. Mar. Biol. Assoc. UK 71, 495–513.
Southward, A.J., Crisp, D.J., 1954. Recent changes in the distribution of the intertidal barnacles *Chthamalus stellatus* and *Balanus balanoides* in the British Isles. J. Anim. Ecol. 23, 163–177.
Southward, A.J., Hawkins, S.J., Burrows, M.T., 1995. Seventy years' observations of changes in distribution and abundance of zooplankton and intertidal organisms in the western English Channel in relation to rising sea temperature. J. Therm. Biol 20 (1–2), 127–155.
Southward, A.J., Langmead, O., Hardman-Mountford, N.J., Aiken, J., Boalch, G.T., Dando, P.R., Genner, M.J., Joint, I., Kendall, M.A., Halliday, N.C., Harris, R.P., Leaper, R., Mieszkowska, N., Pingree, R.D., Richardson, A.J., Sims, D.W., Smith, T., Walne, A.W., Hawkins, S.J., 2005. Long-term oceanographic and ecological research in the western English Channel. Adv. Mar. Biol. 47, 1–105.
Steinke, S., Yu, P.S., Kucera, M., Chen, M.T., 2008. No-analog planktonic foraminiferal faunas in the glacial southern South China Sea: implications for the magnitude of glacial cooling in the western Pacific warm pool. Mar. Micropaleo. 66, 71–90.
Thomas, C.W., Crear, B.J., Hart, P.R., 2000. The effect of temperature on survival, growth, feeding and metabolic activity of the southern rock lobster, *Jasus edwardsii*. Aquaculture 185, 73–84.
Thompson, R., Clark, R., 2008. Is spring starting earlier? Holocene 18, 95–104.
Visser, M., Both, C., 2005. Review. Shifts in phenology due to global climate change: the need for a yardstick. Proc. R. Soc. B. 272, 2561–2569.
Watanuki, Y., Ito, M., Deguchi, T., Minobe, S., 2009. Climate-forced seasonal mismatch between the hatching of rhinoceros auklets and the availability of anchovy. Mar. Ecol. Prog. Ser. 393, 259–271.
Wernberg, T., Thomsen, M.S., Tuya, F., Kendrick, G.A., Staehr, P.A., Toohey, B.D., 2010. Decreasing resilience of kelp beds along a latitudinal temperature gradient: potential implications for a warmer future. Ecol. Lett. 13, 685–694.
Wernberg, T., Thomsen, M.S., Tuya, F., Kendrick, G.A., in press. Biogenic habitat structure of seaweeds change along a latitudinal gradient in ocean temperature. J. Exp. Mar. Biol. Ecol.
Wethey, D.S., 1983. Geographic limits and local zonation: the barnacles *Semibalanus (Balanus)* and *Chthamalus* in New England. Biol. Bull. 165, 330–341.
Wethey, D.S., 1984. Sun and shade mediate competition in the barnacles *Chthamalus* and *Semibalanus*: a field experiment. Biol. Bull. 167, 176–185.
Wethey, D., Woodin, S., 2008. Ecological hindcasting of biogeographic responses to climate change in the European intertidal zone. Hydrobiologia 606, 139–151.
Williams, J.W., Jackson, S.T., 2007. Novel climates, no-analog communities, and ecological surprises. Front. Ecol. Environ. 5, 475–482.
Wonham, M.J., O'Connor, M., Harley, C.D.G., 2005. Positive effects of a dominant invader on introduced and native mudflat species. Mar. Ecol. Prog. Ser. 289, 109–116.
Wood, S.A., Lilley, S.A., Schiel, D.R., Shurin, J.B., 2010. Organismal traits are more important than environment for species interactions in the intertidal zone. Ecol. Lett. 13, 1160–1171.